



VITAMINS & NUTRITION — FULL MIND MAP

Biology Ch. 7.iii · Human Anatomy & Physiology · Hinglish · SSC / BPSC / BSSC / Railway

Topics in this sub-chapter:

1. Nutrition Basics, Probiotics & Nutraceuticals
2. Vitamins — Concept, Solubility & Storage
3. Minerals, Potassium & Antioxidants
4. Vitamin C — Sources, Functions & Storage
5. Vitamin K — Blood Clotting
6. Vitamin D — Sunlight, Forms & Deficiency
7. Vitamin A — Vision, Sources & Storage
8. B-Complex Vitamins & Vitamin B12
9. Vitamin Sources, Names & Deficiency Diseases
10. Proteins, Carbohydrates & Energy
11. Iron, Milk & Dairy Nutrition
12. Protein Nutrition, Micronutrients & Breast Milk
13. Starch, Malnutrition, Fatty Acids & Amino Acids
14. Fats, Oils, Cholesterol & Triglycerides

1. Nutrition Basics, Probiotics & Nutraceuticals

- New tissue growth aur repair ke liye proteins important hain; source cheese ko given options mein protein-rich choice batata hai. Pulses, soybean, groundnut, milk, cheese, eggs aur foodgrains bhi protein sources diye gaye hain. [Q1]
- Probiotics live beneficial microorganisms hain—bacteria aur yeasts—jo gut flora maintain karne, immunity support karne aur milk sugar/lactose digestion mein help kar sakte hain. Source ke mutabik probiotics naturally gut mein bhi present hote hain. [Q2, Q3]
- Nutraceuticals aise food/food-supplement products hain jo simple nutrition ke beyond medicinal/health benefit dete hain. Casimir Funk ko vitamin concept / “vital amines” se credit diya gaya hai; “accessory dietary factors” later vitamins ke naam se jane gaye. [Q4, Q5, Q8]
- Source discovery pair note: McCollum & Marguerite Davis → Vitamin A; James Lind → Vitamin C/scurvy; Paul Müller Vitamin D se associated nahi, DDT insecticidal use se associated hai. [Q6]
- Vitamins organic compounds / vital nutrients hain jo small quantities mein required hote hain aur generally body sufficient amount mein synthesize nahi kar pati. Proper growth ke liye essential hain. [Q7, Q9, Q10]

2. Vitamins — Concept, Solubility & Storage

- Fat-soluble vitamins = A, D, E, K. Water-soluble vitamins = B-complex + C. Source ke multiple questions isi classification ko repeat karte hain. [Q12, Q13, Q14, Q15, Q16, Q17, Q18]
- Liver fat-soluble vitamins A, D, E, K ke saath Vitamin B12 ko bhi store karta hai. [Q11]



EXAM TRAP — Water-Soluble Does Not Always Mean “Not Stored”

- ▶ Source Vitamin C ko body mein store na hone wala batata hai, lekin Vitamin B12 water-soluble hote hue bhi liver mein 3–5 years tak stored reh sakta hai. [Q34, Q63]

3. Minerals, Potassium & Antioxidants

- Micro-minerals / trace elements very small quantities mein required hote hain; source examples mein iron, copper, zinc etc. deta hai. Macronutrient elements mein potassium, sodium, magnesium, calcium, sulphur, phosphorus aur chlorine jaise elements include hain. [Q19, Q20]
- Source apple ko potassium-rich batata hai aur heart rhythm / muscle action se potassium ko associate karta hai. [Q21]

- Antioxidants free radicals ke harmful effects se cells ko protect karte hain; oxidation/free-radical damage ko limit karke health maintenance mein role play karte hain. [Q22, Q23]

4. Vitamin C — Sources, Functions & Storage

- Vitamin C = Ascorbic acid. Citrus fruits, raw fruits, lemon, orange, amla, papaya etc. major sources hain. Source questions mein chilli given vegetables mein maximum, aur context/options ke hisab se amla/orange ko richest/best source select kiya gaya hai. [Q24, Q25, Q26, Q27, Q28, Q29, Q30, Q31]
- Vitamin C healthy skin / normal growth, wound healing aur iron absorption mein helpful hai. Water-soluble hone ke karan source ke mutabik body mein store nahi hota. [Q32, Q33, Q34]

EXAM TRAP — “Richest Vitamin C Source” Depends on Given Options

- ▶ Source kuch questions mein Amla ko richest source batata hai, jabki doosre option-based questions mein Orange select karta hai; vegetable-only question mein Chilli selected hai. Question ke options/context ko follow karo. [Q24, Q27, Q28, Q29, Q31]

5. Vitamin K — Blood Clotting

- Vitamin K blood coagulation/clotting ke liye essential hai; deficiency se slow/impaired coagulation aur uncontrolled bleeding ho sakti hai. Isse anti-hemorrhagic factor bhi kaha gaya hai. [Q35, Q36, Q37, Q38, Q39, Q40]
- Vitamin K anticoagulant poison ke antidote ke roop mein use hota hai. Source intestinal bacteria ko Vitamin K aur Vitamin B12 synthesis se associate karta hai. [Q41, Q42]
- Vitamin K deficiency sterility ka cause nahi; source sterility ko Vitamin E deficiency se associate karta hai. [Q43]

6. Vitamin D — Sunlight, Forms & Deficiency

- Vitamin D ka important source sunlight hai; skin sunlight presence mein Vitamin D banati hai. Cod-liver oil bhi source ke matching questions mein Vitamin D source diya gaya hai. [Q44, Q45, Q46]
- Vitamin D deficiency children mein rickets aur adults mein osteomalacia se associated hai; prolonged deficiency osteoporosis mein contribute kar sakti hai. [Q47]
- Vitamin D forms: D2 = Ergocalciferol, D3 = Cholecalciferol. Source Vitamin D ko hormone-like vitamin / hormone ke roop mein describe karta hai; active form calcitriol hai. [Q48, Q49, Q50, Q51]

EXAM TRAP — Q47 — More Than One Deficiency Outcome

- ▶ Source explanation explicitly kehti hai Vitamin D deficiency Rickets (children) aur Osteomalacia (adults) dono cause karti hai; is question mein “both A and C” source conclusion hai. [Q47]

7. Vitamin A — Vision, Sources & Storage

- Vitamin A normal vision ke liye essential hai; deficiency night blindness / nyctalopia aur severe cases mein xerophthalmia/blindness se linked hai. [Q52, Q58, Q59]
- Vitamin A liver mein stored hota hai. Source-rich food questions carrots, mango/papaya, cod-liver oil etc. ko Vitamin A sources se associate karte hain. [Q53, Q54, Q55, Q56]
- Source ke mutabik human body Vitamin A synthesize nahi karti; diet se lena hota hai. [Q57]

8. B-Complex Vitamins & Vitamin B12

- Vitamin B12 = Cobalamin / Cyanocobalamin aur isme Cobalt present hota hai. Water-soluble hone ke bawajood source B12 ko liver mein approx. 3–5 years stored batata hai. [Q60, Q61, Q62, Q63, Q64]
- Vitamin B1 = Thiamine; deficiency Beri-beri. Source combined deficiency question mein Beri-beri → B1, Rickets → D, Scurvy → C mapping preserve karta hai. [Q65, Q66]

- Source Q67 Vitamin B6/B12/folate intake ko dream quality/recall par possible impact se associate karta hai; selected option B6 tha. [Q67]

9. Vitamin Sources, Names & Deficiency Diseases

- Source matchings: Vitamin K → Alfalfa; D → Cod-liver oil/sunlight; E → Wheat-germ oil; C → Lemon/Orange; B2 (Riboflavin) → eggs, green vegetables, milk/dairy etc. Orange mainly Vitamin C source hai. [Q68, Q69, Q70, Q71, Q85]
- Classic deficiency map: Vitamin A → night blindness/blindness; B1 → Beri-beri; C → Scurvy; D → Rickets; Folic acid (B9) → Anaemia. Vitamin E is not the cause of night blindness. [Q72, Q73, Q74, Q75, Q76, Q77]
- Riboflavin (B2) deficiency → ariboflavinosis, stomatitis/cheilosis etc. Source mismatch questions also distinguish B1/Thiamine, B2/Riboflavin, E/Tocopherol, B12/Cyanocobalamin and D/Calciferol. [Q78, Q79, Q80, Q81, Q82]
- Vitamin E/Tocopherol source Wheat-germ oil aur role antisterility factor; Vitamin K = naphthoquinone/anti-hemorrhagic factor. Thyroxine vitamin nahi, thyroid hormone hai. [Q83, Q84, Q85, Q86]

10. Proteins, Carbohydrates & Energy

- Source banana ko approx. 89 kcal/100 g ke around high-calorie fruit batata hai. Legumes protein-rich hain; soybean/groundnut/gram/pulses jaise foods high-protein sources ke roop mein repeat hote hain. [Q87, Q88, Q89, Q90, Q91, Q92, Q93]
- Rice/starchy foods mainly carbohydrate hain. Wheat bread-making quality gluten se linked hai. Tyrosine protein-rich foods mein amino acid hai; source Q95 ke statement set ko option-based way mein evaluate karta hai. [Q94, Q95, Q96]
- Hard worker ke daily energy requirement ko source ~4000 kcal/day range se associate karta hai; water zero calories deta hai. Diet ka essential constituent water hai; long-duration exercise mein major food fuel source ke mutabik fats hain. [Q97, Q98, Q99, Q100]
- Nitrogen proteins ka characteristic element hai, carbohydrates ka essential constituent nahi. Source body acid-alkali balance ko sodium se associate karta hai; Vitamin C ko body mein stored na hone wala compound batata hai. [Q101, Q102, Q103]

11. Iron, Milk & Dairy Nutrition

- Source fruit/food questions iron-rich sources mein dates/raisins/green leafy vegetables/spinach type options ko highlight karte hain. Muscle contraction ke liye calcium, iron aur sodium ke statement-based combination ko source answer follow karta hai. [Q104, Q105, Q106, Q107, Q108]
- Basmati rice grains cooking par elongate hone ko source higher amylose content se associate karta hai. [Q109]
- Milk ki sweetness Lactose se hoti hai; lactose milk sugar/saccharide hai. Lactobacillus lactic-acid bacteria milk ko sour/curd banane se associated hain. [Q110, Q111, Q112, Q113]
- Cow milk ka yellowish tint carotene se linked hai. Buffalo milk fat source ke mutabik approx. 7–7.5% range; cow milk lower fat range mein diya gaya hai. [Q114, Q115, Q116]
- Milk digestion: rennin young mammals mein milk protein curdling/digestion mein role karta hai; source ke mutabik rennin/lactase production childhood mein decline hoti hai. Casein milk ka phosphoprotein hai aur milk ke white colour se associate kiya gaya hai; milk mein casein/lactalbumin/lactoglobulin jaise proteins milte hain. [Q117, Q118, Q119, Q120, Q121]

EXAM TRAP — Q117 — Source Gives an Age Range, Not One Sharp Age

- ▶ Source explanation: rennin/lactase decline 2–3 years se start ho sakta hai aur most humans mein 6–15 years ke beech complete disappearance bataya gaya hai. Isko single universal age mat samjho. [Q117]

12. Protein Nutrition, Micronutrients & Breast Milk

- Proteins body tissues ke basic structural/building molecules hain; deficiency slow growth se linked hai. Proteins mein Carbon, Hydrogen, Oxygen aur Nitrogen commonly present hote hain. [Q122, Q123, Q124]
- Important element matches: Vitamin B12 → Cobalt; Haemoglobin → Iron; Chlorophyll → Magnesium; Brass/Chalcopyrite → Copper. Vitamin names: B1 → Thiamine, B2 → Riboflavin, B6 → Pyridoxine, B12 → Cobalamin. [Q125, Q126, Q127, Q128]
- Micronutrients small quantities mein required; source Sulphur ko given choices mein “not micronutrient” / macronutrient category mein places. Papaya mainly Vitamin C rich; source Q131 option-context mein milk ko “not natural source of Vitamin A” marks karta hai. [Q129, Q130, Q131]
- Protein denaturation heat, X-rays, heavy-metal salts se ho sakti hai; source infrared rays ko non-denaturing option batata hai. Proteins body building/growth-repair ke liye important hain aur animal + plant sources dono se milte hain. [Q132, Q133]
- Source protein requirement older adults mein potentially higher batata hai; moderately active woman ~45–46 g/day aur lactating mother ~65–70 g/day protein figures deta hai. [Q134, Q135, Q136]
- Milk B-group vitamins, protein, carbohydrate, calcium, potassium etc. provide karta hai. Mother's milk newborn ke liye near-complete diet hai but source iron ko lacking nutrient batata hai; colostrum protein + antibodies rich hai. Breast-fed babies ko better disease resistance aur lower obesity se associate kiya gaya hai. [Q137, Q138, Q139, Q140]
- Milk source ke mutabik Vitamin C ka poor source hai. [Q141]

13. Starch, Malnutrition, Fatty Acids & Amino Acids

- Starch glucose units ka polymeric carbohydrate/polysaccharide hai. Starch iodine solution ke saath blue-black colour deta hai; cellulose does not. Source statement question mein starch/cellulose differences ko isi framework se test karta hai. [Q142, Q143]
- Malnutrition undernutrition, overnutrition aur imbalanced nutrition—teeno situations se ho sakta hai. [Q144]
- Essential fatty acids diet ka part hone chahiye kyunki body sab fatty acids synthesize nahi kar sakti. Linseed/flaxseed oil alpha-linolenic acid (plant omega-3) ka rich source diya gaya hai. [Q145, Q146, Q147]
- Vitamins micronutrient amounts mein health ke liye required hain, lekin harmful bacteria ko directly vitamins nahi maarte; source white blood cells ko defensive role deta hai. [Q148]
- Glycine, serine aur tyrosine source mein non-essential amino acids hain; essential amino acids body synthesize nahi kar sakti. Complete proteins all essential amino acids provide karte hain; source eggs ko best/high biological value source batata hai. [Q149, Q150, Q151]

EXAM TRAP — Q150 — No Single Option Correct in Source

- ▶ Source explanation ke mutabik given choices mein rice ko chhodkar multiple foods all essential amino acids provide karte hain; isliye source explicitly kehta hai “no single option is correct.” [Q150]

14. Fats, Oils, Cholesterol & Triglycerides

- Subcutaneous fat thermal insulation / heat-loss barrier ke roop mein act karta hai. Fats per gram sabse zyada energy dete hain (~9.3 kcal/g source), jabki protein aur carbohydrate ~4 kcal/g. [Q152, Q153, Q154]
- Maximum energy-providing diet group mein carbohydrate + fat + protein included hain; vitamins/minerals/water significant calories nahi dete. [Q155]

 EXAM TRAP — Q156 — Source Wording: “Living Material”

► Source pehle water ko living beings mein most abundant substance maanta hai, phir “organic compounds / living materials” ke context mein protein ko most abundant body organic material batata hai. Q156 ka source answer isi wording ko follow karta hai. [Q156]

- Source cooking oil ko vanaspati ghee se better isliye batata hai kyunki unsaturated fats heart-healthier choices hain. Sunflower oil unsaturated fatty acids rich aur heart patients ke liye suitable option diya gaya hai. [Q157, Q158, Q159]
- Unsaturated fats double bonds ke karan saturated fats se more reactive hote hain. Fish ko polyunsaturated fatty acids ke karan healthier choice bataya gaya hai. [Q160, Q161]
- Trans fats HDL (“good cholesterol”) ko lower aur LDL ko raise kar sakte hain; source trans-fat-free claim ko hydrogenated oils se bachav aur cardiovascular-health context se relate karta hai. [Q162, Q163]
- Cholesterol lipid/steroid-family ka sterol / fatty alcohol hai; abnormal high level arterial blockage/atherosclerosis se linked hai. Triglyceride glycerol + three fatty acids ka ester aur body/vegetable fat ka major constituent hai. [Q164, Q165, Q166, Q167]